

Machine Learning Assignment 2

Obesity Risk Classification & Deployment

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1. Submission Links

* GitHub Repository:
[https://github.com/2025aa05090-nitin/ML_Assignment_2](https://github.com/2025aa05090-nitin/ML_Assignment_2)

* Streamlit App:
<https://mlassignment2-2025aa05090.streamlit.app/>

2. Execution Screenshot

Screenshot of assignment execution on BITS Virtual Lab:



Screenshot of assignment execution on BITS Virtual Lab

3. README Content

Problem statement

Build and evaluate multiple classification models on a public dataset, then deploy a Streamlit app that allows model selection, metric display, and prediction on uploaded test data.

Dataset description

Dataset: Obesity Risk Classification (Kaggle).

- Source: Kaggle - Obesity Risk Classification Dataset
- Dataset Link: <https://www.kaggle.com/datasets/fernandoramirez/obesity-risk-classification>
- Rows: 600
- Features: 17 input features (numerical + categorical)
- Target: NObeyesdad (7 classes: Normal_Weight, Overweight_Level_I, Overweight_Level_II, Obesity_Type_I, Obesity_Type_II, Obesity_Type_III, Insufficient_Weight)
- Source file: data/obesity_risk.csv

Models used (Comparison Table)

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.7417	0.9511	0.7250	0.7417	0.7123	0.6651
Decision Tree	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
K-Nearest Neighbors	0.5000	0.7986	0.4362	0.5000	0.4481	0.3347
Naive Bayes	0.8583	0.9853	0.8869	0.8583	0.8633	0.8213
Random Forest	0.9000	0.9965	0.8580	0.9000	0.8742	0.8738
XGBoost	0.9917	1.0000	0.9944	0.9917	0.9921	0.9894

Observations

ML Model Name	Observation about model performance
Logistic Regression	Decent baseline (74%) but struggles with non-linear relationships in biological data compared to tree-based models.
Decision Tree	**Perfect Performance (100%)** . The model mastered the deterministic rules ($BMI = Weight/Height^2$) underlying the dataset labels.
K-Nearest Neighbors	Weakest performer (50%). Likely impacted by the high dimensionality (after encoding) and scale differences, despite feature scaling.
Naive Bayes	Surprisingly strong (86%), though the independence assumption is theoretically violated by correlated features like Weight and BMI.
Random Forest	Strong performance (90%) but slightly overfitted compared to the single Decision Tree, possibly due to noise in the bagging process or hyperparameter settings.
XGBoost	**Excellent (99.2%)** . Nearly perfect. It captures complex patterns effectively and is the most robust alternative to the Decision Tree.

Deployment

The application is deployed on Streamlit Cloud.

- Features:
 - EDA Tab: Visualizes target distribution and correlations.
 - Model Comparison: Interactive leaderboards.
 - Live Predictions: Upload CSV to run all 6 models simultaneously.